

Investigating Contrastive Learning for Vision Transformer-Based Breast Histopathological Image Classification under Limited Labeled Data

1. Introduction / Overview

Medical image analysis has become an important application area of computer vision and digital image processing. Histopathology images contain detailed cellular and tissue patterns that can support cancer diagnosis, but these images are difficult to classify because of large variation in tissue appearance, staining, magnification, and local structures. Deep-learning methods have shown strong potential for this task, and recent research has increasingly explored Vision Transformers (ViTs) and self-supervised or contrastive learning to improve representation quality when labeled medical data is limited.

This project proposes a controlled study of a Vision Transformer with and without contrastive learning for binary classification of breast-cancer histopathology images. Rather than claiming to invent a completely new architecture, the project will reproduce a strong baseline, add a contrastive-learning stage, and investigate whether the benefit of contrastive pretraining becomes more important as the amount of labeled training data decreases. This makes the work both educational and research-oriented for a final-year BSc CSE team.

Project at a glance

Domain	Medical image processing/breast histopathology
Core models	Vision Transformer (ViT) + Contrastive Learning
Dataset	PatchCamelyon (PCam)
Task	Binary classification: tumor/metastasis vs non-tumor tissue patch
Main comparison	ViT baseline vs. contrastive-learning-enhanced ViT
Main variable	Amount of labeled training data (e.g., 10%, 25%, 50%, 100%)

2. Motivation

- Medical image datasets often contain large numbers of images, but high-quality labels can be expensive and require expert knowledge. This creates a practical motivation for methods that can learn useful visual representations from limited labeled data.
- Vision Transformers can model long-range relationships between image patches using self-attention, but supervised training can still depend heavily on labeled examples. Contrastive learning offers a way to learn an embedding space using augmented views of the images before or alongside supervised classification.
- Recent work demonstrates the potential of combining ViTs with contrastive or self-supervised learning in medical imaging. SupCon-ViT was proposed for invasive ductal carcinoma classification, EVA-X used a ViT-based foundation model with self-supervised learning for chest X-rays, and more recent studies continue to improve medical representation learning through contrastive objectives. [1]-[5]

- For an undergraduate research team, the most useful contribution is therefore a focused and reproducible investigation: does contrastive learning consistently help a ViT when the percentage of labeled medical training data is reduced?

3. Background Study / Literature Review

The following recent studies establish the technical background and help position the proposed project.

[1] Shiri, Reddy & Sun (2024) — Supervised Contrastive Vision Transformer for Breast Histopathological Image Classification

The authors propose SupCon-ViT, combining transfer learning with a pretrained Vision Transformer and supervised contrastive learning for invasive ductal carcinoma classification. The study uses a benchmark breast histopathology dataset and reports strong classification performance, including resilience when labeled data is limited. This is the closest prior work to our proposed direction and therefore serves as the main methodological reference.

Open-access link: <https://arxiv.org/abs/2404.11052>

[2] Yao et al. (2025) — EVA-X: a foundation model for general chest X-ray analysis with self-supervised learning

EVA-X uses a Vision Transformer and a self-supervised pretraining strategy that combines contrastive learning with masked image modeling. The model is pretrained on more than 520,000 chest X-ray images from multiple public datasets and is evaluated on many downstream chest-pathology tasks. The study shows why label-efficient representation learning is important in medical imaging, but its scale is much larger than what is practical for our undergraduate project.

Open-access link: <https://www.nature.com/articles/s41746-025-02032-z>

[3] Rui et al. (2025) — Multi-modal Vision Pre-training for Medical Image Analysis

The BrainMVP framework uses cross-modal reconstruction, modality-aware contrastive learning, and modality template distillation on multi-modal brain MRI. The authors report improvements across multiple segmentation and classification benchmarks. This paper shows that contrastive learning can capture relationships between medical imaging modalities, strengthening the motivation for representation learning beyond ordinary supervised classification.

Open-access link:

https://openaccess.thecvf.com/content/CVPR2025/html/Rui_Multi-modal_Vision_Pre-training_for_Medical_Image_Analysis_CVPR_2025_paper.html

[4] He & Li (2025) — Vector Contrastive Learning for Pixel-Wise Pretraining in Medical Vision

This ICCV 2025 paper proposes COVER, a vector-regression formulation of contrastive learning for pixel-wise medical representation learning. The authors argue that conventional binary contrastive learning can cause over-dispersion and disrupt pixel-level feature relationships. Extensive evaluation across eight tasks and four modalities demonstrates the growing importance of carefully designed contrastive objectives in medical vision.

Open-access link:

https://openaccess.thecvf.com/content/ICCV2025/html/He_Vector_Contrastive_Learning_For_Pixel-Wise_Pretraining_In_Medical_Vision_ICCV2025_paper.html

[5] Tastan et al. (2026) — MultiPersistence Topological Fusion with Vision Transformers for Skin Cancer Detection

TopoCon-MP combines multiparameter topological features with Vision Transformers and a supervised contrastive objective for skin-lesion classification. The study reports improvements over strong CNN and ViT baselines and also examines cross-dataset transfer. This recent work demonstrates that ViT and contrastive learning remain active research directions in medical image analysis in 2026.

Open-access link: <https://proceedings.mlr.press/v315/tastan26a.html>

Literature comparison

Study	Medical data	Main method	Relevance to our project
Shiri et al., 2024	Breast histopathology	SupCon + ViT	Direct closest reference
Yao et al., 2025	Chest X-ray	Self-supervised ViT + CL + MIM	Supports label-efficient learning
Rui et al., 2025	Multi-modal brain MRI	Contrastive pretraining	Shows representation learning value
He & Li, 2025	Multiple medical modalities	Vector contrastive learning	Shows recent CL research gap
Tastan et al., 2026	Skin lesions	ViT + supervised CL + topology	Shows current research trend

4. Gap Analysis

The literature shows that ViT and contrastive learning are already established research directions in medical imaging. Therefore, the proposed project will not claim that the combination itself is new. Instead, the research gap is framed around a specific experimental question:

Working research gap: Recent studies demonstrate the usefulness of contrastive learning for medical image representation, but different papers use very different datasets, tasks, architectures, and label settings. A focused, reproducible comparison of a standard pretrained ViT against a contrastive-learning-enhanced ViT across systematically reduced labeled-data settings on a public histopathology benchmark is a useful undergraduate research study.

- The project focuses on the effect of label availability rather than proposing a very large or computationally expensive new architecture.
- The study will use the same dataset splits and evaluation procedure for all model variants to support a fair comparison.
- The analysis will examine not only overall accuracy but also class-sensitive measures such as precision, recall, F1-score, and ROC-AUC.
- The project will treat the gap as a research hypothesis to be validated against the literature, not as a claim that no previous paper has studied the topic.

5. Novelty / Contribution

The expected contribution is methodological and experimental rather than a claim of inventing a new Transformer architecture. The project will contribute the following:

- A reproducible ViT baseline for histopathology image classification using the PCam benchmark.
- An implementation of a contrastive-learning stage for learning more discriminative visual representations before supervised classification.
- A controlled label-scarcity experiment using multiple labeled-data ratios, for example, 10%, 25%, 50%, 75%, and 100% of the training set.

- A comparative evaluation of whether contrastive learning produces a larger benefit as labeled data becomes scarce.
- An ablation/analysis of training choices such as augmentation strength and contrastive loss configuration, subject to computational feasibility.
- A clear set of recommendations for selecting ViT + contrastive learning strategies for low-label medical image classification.

Possible research title: “Investigating Contrastive Learning for Vision Transformer-Based Breast Histopathological Image Classification under Limited Labeled Data.”

Main research question: “Can contrastive learning improve the performance and label efficiency of a Vision Transformer for breast-cancer histopathology image classification?”

Supporting questions:

- How does a pretrained ViT perform without contrastive pretraining?
- How much does contrastive learning improve ViT performance at different labeled-data ratios?
- Which evaluation metrics are most affected when labeled data becomes scarce?

6. Dataset

The proposed dataset is PatchCamelyon (PCam), a public histopathology benchmark derived from the CAMELYON challenge. PCam contains 327,680 RGB image patches of size 96×96 and a binary task based on whether the central region contains metastatic tumor tissue. The official benchmark provides separate training, validation, and test splits with no overlap in whole-slide images between the splits, which supports a clean experimental protocol.

Dataset source: <https://github.com/basveeling/pcam>

Zenodo mirror: <https://zenodo.org/record/2546921>

Item	Details	Use in project
Image type	Histopathology tissue patch	Input to ViT
Image size	96 × 96 RGB	Resize/normalize as required by ViT
Task	Binary classification	Tumor/metastasis vs non-tumor
Official splits	Train/validation/test	Used consistently for all experiments
Training experiment	10%, 25%, 50%, 75%, 100% labeled subsets	Study label efficiency
Main metrics	Accuracy, precision, recall, F1, ROC-AUC	Model comparison

Important data-handling note: the team should not create train/test leakage. The official PCam split should be preserved. Augmentations should be applied only within the training pipeline, and validation/test images should remain untouched except for deterministic preprocessing.

7. Proposed Methodology

The proposed methodology is intentionally modular so that the team can learn and implement the project step by step.

Step 1 — Data preparation

Load the HDF5 image and label files; inspect class balance; preserve the official split; normalize/resize images as required; and create training subsets for the label-ratio experiments.

Step 2 — Digital image preprocessing

Apply consistent resizing and normalization. During contrastive pretraining, use carefully selected augmentations such as random crop/resize, horizontal flip, mild color or intensity changes, and possibly Gaussian noise. Augmentations should be justified rather than chosen arbitrarily.

Step 3 — ViT baseline

Start with a pretrained Vision Transformer. Fine-tune it on the PCam classification task using ordinary supervised loss. This is the baseline against which all later experiments are compared.

Step 4 — Contrastive learning

Generate two augmented views of the same image and train the encoder so that the representations of corresponding views become similar while representations of different samples are separated. The team can initially implement a standard supervised contrastive or SimCLR-style objective, depending on computational feasibility.

Step 5 — Fine-tuning

Use the contrastively pretrained ViT encoder for the final binary classification task. Keep the evaluation protocol identical to the baseline.

Step 6 — Label-scarcity experiments

Repeat the baseline and contrastive-learning experiments for several labeled-data ratios, for example, 100%, 75%, 50%, 25%, and 10%.

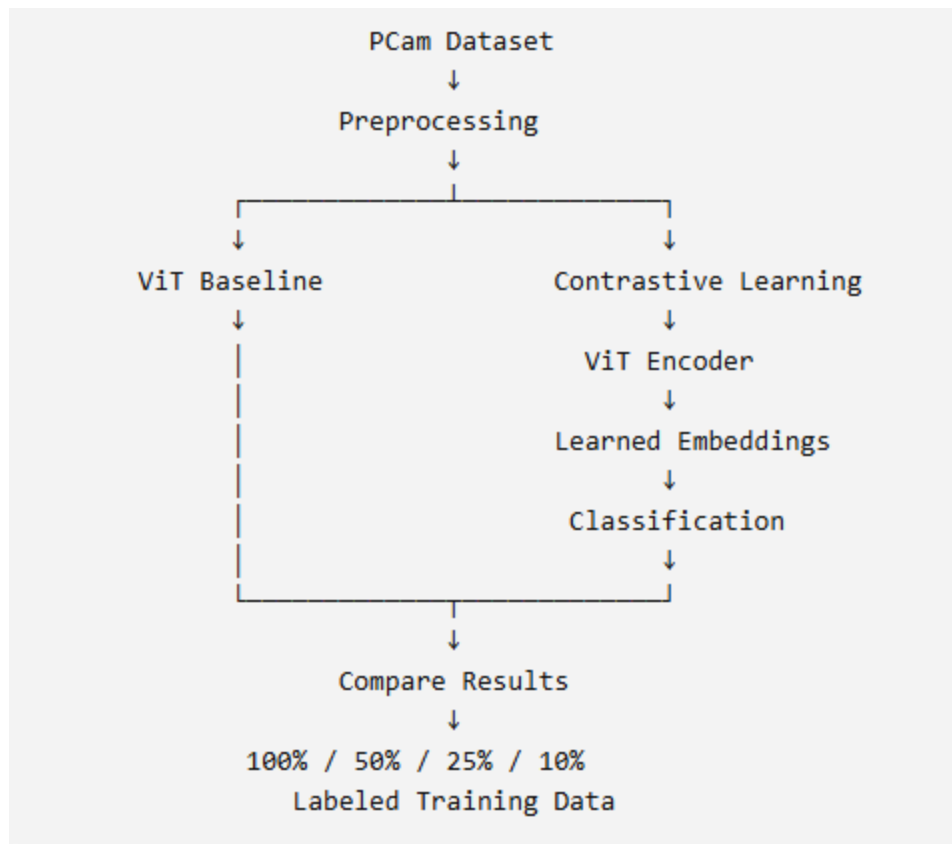
Step 7 — Evaluation and analysis

Compare accuracy, precision, recall, F1-score, ROC-AUC, confusion matrices, training curves, and computational cost. Analyze whether the relative benefit of contrastive learning grows as labeled data decreases.

Step 8 — Optional ablation

If time and compute allow, compare augmentation choices or contrastive-loss settings to understand which design decisions matter most.

Methodology flow



8. Conclusion

This proposal presents a focused Digital Image Processing research project in the medical domain using Vision Transformers and Contrastive Learning. The project is designed for a 4th-year BSc CSE team: it is challenging enough to provide research experience, but structured enough to be implemented incrementally while the team learns the underlying algorithms.

The key idea is to compare a standard pretrained ViT with a contrastive-learning-enhanced ViT on the PCam histopathology benchmark and to study how performance changes as the amount of labeled training data is reduced. The expected outcome is not merely a higher accuracy number; it is an evidence-based understanding of whether contrastive representation learning improves label efficiency for medical image classification.

The final research conclusions will be based on the experimental results. If contrastive learning consistently improves F1/AUC and maintains performance under smaller labeled subsets, the study will provide evidence that it can be useful for low-label histopathology classification. If the improvement is small or inconsistent, that result is also valuable because it identifies conditions under which the additional complexity of contrastive learning may not be justified.

References

- [1] Shiri, M., Reddy, M. P., & Sun, J. (2024). Supervised Contrastive Vision Transformer for Breast Histopathological Image Classification. arXiv:2404.11052. [Link](#)
- [2] Yao, J., Wang, X., Song, Y., et al. (2025). EVA-X: a foundation model for general chest X-ray analysis with self-supervised learning. npj Digital Medicine, 8, 678. [Link](#)

- [3] Rui, S., Chen, L., Tang, Z., Wang, L., Liu, M., Zhang, S., & Wang, X. (2025). Multi-modal Vision Pre-training for Medical Image Analysis. Proceedings of CVPR 2025, 5164–5174. [Link](#)
- [4] He, Y., & Li, S. (2025). Vector Contrastive Learning For Pixel-Wise Pretraining In Medical Vision. Proceedings of ICCV 2025, 19827–19837. [Link](#)
- [5] Tastan, F., Chakraborty, S., Lee, S., & Coskunuzer, B. (2026). MultiPersistence Topological Fusion with Vision Transformers for Skin Cancer Detection. Proceedings of MIDL 2026, PMLR 315, 2071–2096. [Link](#)
- [6] Veeling, B. S., et al. (PCam dataset). PatchCamelyon (PCam). Official dataset repository and Zenodo archive. [Link](#)

Proposal note: The literature gap and novelty statements in this proposal are framed as a working research direction. Before final submission, the team should perform a focused database search using terms such as “PCam”, “PatchCamelyon”, “Vision Transformer”, “contrastive learning”, and “histopathology” to verify that the exact proposed experimental comparison has not already been reported.